What a Remaining-Useful-Life Interval Can Prove: Coverage, Width, and the Calibration Object

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***Abstract*—Prediction intervals for remaining useful life (RUL) are reported as a coverage rate close to nominal and a width as small as possible. This paper asks how much of that pair is information and how much is convention, using 653 C-MAPSS test engines and 2 587 N-CMAPSS queries. Three conventions are priced. Scoring coverage on the integer grid inflates PICP by $+0.011$ to $+0.044$ at the 90 % level and $+0.074$ to $+0.120$ at 50 %, independent of the model; randomised PIT removes the shift. Reporting the *central* interval of a predictive distribution rather than the *shortest* interval of the same mass costs 10.4 % of the width at the 90 % level and 33 % at 50 %. Calibrating a cross-validation fold model while deploying an ensemble — the standard CV+ construction — costs a further 11 %. A matched-object construction reaches 32.4 cycles against the 30.9 of a 66-parameter parametric layer, with $|\Delta| = 0.014$ across fleets where the layer ranges 0.842–0.932; a leave-one-out ablation finds every ingredient non-negative, variance normalisation the largest ($+5.0$ Winkler points). An uncalibrated fifteen-member deep ensemble covers 0.485–0.553 at nominal 0.90, and conformalised quantile regression 0.883–0.919 at 14–22 % more width. With 11–82 engines per RUL bin the smallest detectable coverage deviation is 0.14–0.44, and 149 engines per bin are needed to detect 0.10. On a second backbone the width ordering replicates on all four fleets while coverage moves by up to 0.13: at this scale coverage measures the predictor–population pair, not the interval rule.**

***Index Terms*—Remaining useful life, uncertainty quantification, conformal prediction, calibration, coverage, measurement uncertainty, prognostics and health management.**

I. Introduction

A remaining-useful-life (RUL) predictor that returns a single number cannot be used to schedule maintenance: the planner needs to know how wrong the number may be. The literature has therefore converged on interval-valued prognostics [17], [18], reported through two statistics — the fraction of engines whose true RUL falls inside the interval (PICP), and the mean interval width. The implicit claim is that the first is at least the nominal level and the second is as small as the method can make it.

Recent work in this journal has advanced that programme considerably. CRULP combines conformalised quantile regression with cross-conformal prediction to turn a point predictor into an interval with a distribution-free guarantee [1], and related studies extend conformal RUL to covariate shift [2], to missing and faulty sensor data [3], and to IoT predictive maintenance [4]. The methodological question these papers raise is no longer *whether* a guarantee can be attached to an RUL interval. It can. The question is what a specific experiment on a specific benchmark can *show* about the interval it reports.

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A parallel literature improves the predictor rather than the interval: evidential and domain-adaptive formulations that carry their own uncertainty [5], bearing prognostics that pair a degradation indicator with a Bayesian network [6], model-based degradation models that derive the first-hitting-time density analytically [7], architecture work on the turbofan benchmarks [8], and conformal active learning that uses the interval width to choose what to label [10]; [9] surveys the health-indicator branch. Two features of that body of work matter here. Most of it reports either a predictive density or an interval, and the convention that converts one into the other is rarely stated, although it moves the reported score by up to 0.12 (Section II). And the uncertainty is usually validated by a coverage number computed on the same benchmark, which at these sample sizes cannot resolve differences below 0.14 (Section II).

This paper takes that question literally and answers it with measurements rather than arguments. We hold the backbone fixed (a 41 858-parameter model, five folds, three seeds), hold the test engines fixed (653 engines for C-MAPSS, 2 587 queries over 39 units for N-CMAPSS), and vary only the construction of the interval. Seven results follow, and each is a number rather than a preference.

1) **Coverage is measured with a biased instrument.** RUL is a lattice-valued target capped at 125 cycles. Reporting an interval by its integer endpoints counts a hit whenever the endpoints straddle the truth, which adds the mass that lies between the endpoint and the quantile it was meant to represent. The bias is $+0.011$ to $+0.044$ at nominal 0.90 and $+0.074$ to $+0.120$ at nominal 0.50 on the four fleets, always positive, and has nothing to do with the predictor. Randomised PIT removes it.
2) **The benchmarks cannot resolve the differences they are used to rank.** With 11–82 engines in an RUL bin, a binomial test at family-wise 5 % detects only deviations of 0.14–0.44 at 80 % power. Detecting a 10-point deviation needs 149 engines per bin, and a 5-point deviation needs 516. At the fleet level the 95 % band on a coverage of 0.90 is ± 0.060 ($n = 93$) to ± 0.038 ($n = 235$), which is of the same order as the differences between methods that published tables rank.
3) **Two construction choices each cost about a tenth of the width.** The central interval of a predictive distribution is not the shortest interval of the same mass; on the frozen layer the shortest-interval rule saves 10.4 % of the width at identical coverage, and 33 % at the 50 % level, so the convention cost is monotone in the nominal level (Table III). Calibrating the fold model rather than the

deployed ensemble costs 11 % in the CV+ construction.

4) **A matched construction closes the gap to a parametric layer without its assumptions.** Calibrating on residuals of the object that is actually deployed, with signed scores and a free tail allocation, gives 32.4 cycles against the layer's 30.9 on the same engines, with coverage 0.897–0.925 against the layer's 0.842–0.932.
5) **Uncalibrated deep ensembles are the largest error in the comparison.** A fifteen-member ensemble reported as $\mu \pm z_{0.95}\sigma$ covers 0.485–0.553 at nominal 0.90. Conformalising a quantile regression on the same engines repairs most of that (0.883–0.919) but is 14–22 % wider than the matched construction.
6) **Each ingredient is priced, and none is negative.** A leave-one-out ablation at nominal 0.90 makes variance normalisation the largest contributor (+5.0 Winkler points when removed, CI excluding zero on three of four fleets) and object matching second (+4.3 and +4.2 on the two fleets where it matters), and finds no removal significantly better than the full construction on any fleet (Table IV). Proposition 1 shows that the tail allocation cannot invalidate the interval and that the equal-tailed convention is optimal only for symmetric score laws, which a capped RUL law is not.
7) **The width ordering survives a change of backbone; the coverage level does not.** Recomputing every construction on a second backbone — a 2.9 times larger LSTM, 10–21 % worse in dev RMSE — reproduces the width ordering on all four fleets (the matched construction is 9–18 % narrower than CV+ on three, 5.4 % wider on the fourth) while coverage moves by up to 0.13. Coverage at this scale is a property of the predictor–population pair, which is why the paper reports width beside it rather than instead of it.

The remainder of the paper is organised as follows. Section II formalises what a coverage number can resolve. Section III defines the constructions and their guarantees. Section IV fixes the experimental protocol. Section V reports results on C-MAPSS, N-CMAPSS and a controlled simulation. Section VI discusses what should be reported, and Section VII concludes.

II. What a Coverage Number Can Resolve

A. Two ways to score the same distribution

Let F_i be a predictive distribution over the integer grid $\{0, \dots, K\}$, $K = 125$, and let Y_i be the realised RUL. An interval $[L_i, U_i]$ covers when $L_i \le Y_i \le U_i$. Two conventions are in use.

Integer endpoints. L_i and U_i are integers, and coverage is the empirical frequency of $L_i \le Y_i \le U_i$.

Randomised PIT. Draw $V_i \sim \mathrm{U}(F_i(Y_i - 1), F_i(Y_i))$ and count $V_i \in [(1-\tau)/2, (1+\tau)/2]$. This is the probability integral transform evaluated inside the atom that contains the observation, and it is unbiased for the nominal level of a *continuous* interval on the same distribution.

The two agree only when the pmf is smooth relative to the grid. For a RUL distribution that is censored at K — where 17 % of the C-MAPSS engines sit exactly at 125 — the endpoint rule adds the atom and part of the neighbouring bin to every interval whose endpoint lands there. Writing b_i for that surplus mass,

$$\widehat{\mathrm{PICP}}_{\mathrm{int}} - \widehat{\mathrm{PICP}}_{\mathrm{PIT}} = \frac{1}{n}\sum_i b_i \;\ge\; 0, \tag{1}$$

so the bias is one-sided and its size is set by how concentrated the pmf is near the endpoints. Table VI reports it: 0.011–0.044 at 0.90 and 0.074–0.120 at 0.50. The layer discussed in Section V would be reported as well calibrated (0.886–0.948) under the integer rule and as under-covering (0.842–0.932) under randomised PIT. Both numbers describe the same object.

B. What n engines can detect

A coverage claim is a hypothesis test. For a bin holding n engines whose true coverage is $p_0 = \tau$, the exact two-sided binomial test at level α rejects when the count falls outside its acceptance region. The smallest $p_1 < \tau$ detected with power $1 - \beta$ is

$$\begin{aligned} \delta^\star(n) &= \tau - \max\{p : \Pr[\mathrm{Bin}(n,p) \le c_\alpha] \ge 1-\beta\}, \\ c_\alpha &= \lfloor \mathrm{Bin}^{-1}(\alpha/2; n, \tau) \rfloor. \end{aligned} \tag{2}$$

The quantity depends only on how many engines fall in the bin and on the nominal level; it is independent of the predictor, the backbone and the interval construction. It is therefore a property of the *benchmark*, not of the method, and it bounds what any method comparison on that benchmark can conclude. Table VII evaluates it at the five RUL bins used in this paper, with a Bonferroni correction over the five bins.

C. Paired comparisons

Two intervals scored on the same engines are paired, so the relevant noise is the discordance count rather than the marginal standard error. We report McNemar's exact test for this. It matters because a difference of 0.03 in marginal coverage can be produced by 3 engines out of 100, while the paired version of the same difference may be significant or not depending on whether the methods err on the same engines.

III. Constructing the Interval

A. Scores

Let $\hat{y}_i$ be the point prediction of the deployed model and $R_i = y_i - \hat{y}_i$ a signed residual. Two score families are standard.

Equal length. Use $s_i = |R_i|$, take q the $\lceil (1-\alpha)(n+1) \rceil$-th smallest score, and report $[\hat{y} - q, \hat{y} + q]$. The interval is symmetric about the prediction by construction.

Conformalised quantile regression [15] learns the two quantiles with a pinball loss instead of taking them from a residual sample; it is the construction used by the closest prior work [1], and the two are equivalent when the quantile model is well specified. We use the residual form because it isolates the effect of the score.

Free tails. Use the signed residuals directly and report $[\hat{y}+q_\gamma, \hat{y}+q_{1-\alpha+\gamma}]$ with $\gamma \in [0,\alpha]$. Setting $\gamma=\alpha/2$ gives equal-tailed probability; setting γ to minimise $q_{1-\alpha+\gamma}-q_\gamma$ gives the shortest interval of mass $1-\alpha$ in the family.

Proposition 1 (The tail allocation is free, and one allocation is shortest)**.** *Let $S_1,\dots,S_m$ be exchangeable scores, S a further score from the same law, and $s_{(1)}\le\cdots\le s_{(m)}$ the ordered calibration scores with $s_{(0)}=-\infty$ and $s_{(m+1)}=+\infty$. For $\gamma\in[0,\alpha]$ set*

$$L_\gamma=\hat{y}+s_{(\lfloor(m+1)\gamma\rfloor)},\qquad U_\gamma=\hat{y}+s_{(\lceil(m+1)(1-\alpha+\gamma)\rceil)}. \quad (3)$$

Then

(i) $\Pr[Y\in[L_\gamma,U_\gamma]]\ge 1-\alpha$ *for every $\gamma\in[0,\alpha]$, so the allocation cannot invalidate the interval;*

(ii) $U_\gamma-L_\gamma$ *is minimised, up to one rank, by the narrowest window of $\lceil(m+1)(1-\alpha)\rceil$ consecutive order statistics, and in the population limit the minimiser is characterised by $f(L)=f(U)$;*

(iii) *the equal-tailed convention $\gamma=\alpha/2$ is optimal only if $f(q_{\alpha/2})=f(q_{1-\alpha/2})$, which fails for any skewed score law, and fails by construction for a capped RUL law, whose upper quantile is the atom at the cap.*

Part (i) is the reason γ can be treated as a free parameter rather than a modelling choice, and it is proved in the Appendix. Part (ii) says the free-tail family contains the shortest interval, so the equal-tailed convention pays for a symmetry the data need not have. Part (iii) is the measurement target of this paper: whether the price is large is an empirical question, and Section V reports both signs — the free tail wins on two fleets and loses on two, which is exactly what a convention that is not a theorem should do.

B. Shortest versus central

For a predictive distribution rather than a score sample — the case of the parametric layer here, and of the analytically derived first-hitting-time densities in [7] — the analogue of the choice is between the central interval, defined by equal tail mass, and the shortest interval of mass τ,

$$[L_i,U_i]=\arg\min_{a\le b}(b-a)\ \text{ s.t. }\ F_i(b)-F_i(a^-)\ge\tau. \quad (4)$$

The two coincide only for symmetric unimodal pmfs. For a censored RUL pmf they do not, and the gap is a pure convention cost: the coverage of the pair is identical because both contain mass τ of the same distribution.

C. Which object is calibrated?

Split conformal [12] with a calibration set of size m guarantees $\Pr[Y\in[L,U]]\ge 1-\alpha$ for exchangeable data, with the finite-sample correction used above. CV+ [13] extends the guarantee to the case where the predictor must be trained on the same data that calibrates it: for each calibration point i held out by fold $k(i)$, the endpoint set is $\{\hat{y}_{-k(i)}(x)+R_i\}$, and the interval is a quantile of that set. Barber et al. prove coverage $\ge 1-2\alpha$ for this construction.

The construction calibrates the object $\hat{y}_{-k}$, a model trained on a $(1-1/K)$ fraction of the data. If what is deployed is the average of the K fold models — the natural choice, and the one used by every ensemble in this paper — then the calibrated object and the deployed object differ by exactly the fold spread. The interval inherits that spread as a fixed width penalty in cycles, which is invisible at $\tau=0.90$ and dominant at $\tau=0.50$.

Matched object. With repeated cross-fitting the mismatch can be removed at no extra cost. If the same partition is used for S seeds, engine i in fold k is excluded from S models, one per seed, and their average is an ensemble that has never seen i. Calibrating on the residuals of that ensemble and centring on the full $K\times S$ ensemble gives a score whose law matches the deployed object up to the ensemble size (S versus KS members).

We call the resulting construction EAC (ensemble-aware conformal): matched-object residuals, signed scores, optional variance normalisation, and a tail allocation γ chosen to minimise the empirical width. Each component is ablated in Section V.

D. Guarantees and what they do not cover

For exchangeable calibration and deployment data, split conformal gives $\Pr[Y\in[L,U]]\ge 1-\alpha$ for any fixed γ (Proposition 1), so the tail allocation cannot invalidate the guarantee. Choosing γ from the same sample is a selection, and the simulation prices it at 0.0016 of coverage at $m=2000$ (nominal 0.90) and 0.009 at $m=200$ (nominal 0.50); a five-fold choice inside the calibration set, scored by the out-of-fold interval score, returns 0.9005 at $m=2000$, within 0.001 of the fixed-γ reference (Section V-H). The guarantee is marginal: it says nothing about coverage conditional on the predicted RUL, and Section II shows that on these benchmarks nothing could.

IV. Protocol

A. Data

C-MAPSS [19]: four fleets (FD001–FD004), 100 training engines each, and 93/235/97/228 test engines, 653 in total. Each test engine contributes one query, taken from its last observable 50-cycle window, which is the deployment query a maintenance system would issue. FD002 and FD004 mix six operating conditions.

N-CMAPSS [21]: nine fleets (DS01–DS08c, DS08d excluded on data-integrity grounds), 60 development and 39 test units, 2 587 cycle-level test queries. The RUL cap is not active in this dataset (maximum true RUL 50–90), so the censoring convention that dominates C-MAPSS is absent; this makes it a useful contrast rather than a replication.

B. Backbone and folds

The C-MAPSS backbone has 41 858 parameters; it is trained with five engine-level folds and three seeds per fold, giving 15 models per fleet. Development predictions are out-of-fold and one row per engine, built with the same feature pipeline as the test predictions. The N-CMAPSS backbone is the 54-64-32-1

network frozen in the earlier protocol, with leave-one-unit-out development predictions and three seeds.

Two artefact checks precede every result. The 15 fold models' test predictions average to the published ensemble prediction within 2.1×10^{-7}; the test target vector agrees with the frozen layer's target on every engine.

C. Metrics and statistics

Coverage is reported twice: randomised PIT for distributions, plain frequency for intervals. Width is the mean of $U - L$ in cycles. The interval score of Winkler [22], one of the standard prognostic metrics [20], with $\alpha = 0.10$ is reported as a sharpness measure that penalises misses. Uncertainty on a coverage is the Wilson interval [23]; uncertainty on a width, and on a difference of widths, is a bootstrap over engines (2 000 resamples), because the engine is the independent unit and resampling windows would understate the interval by the design effect.

D. Simulation

The benchmark cannot check nominal coverage, so a simulation with known truth does. The generator draws a degradation path $Y_0 = 125(1 - u)$, $u \sim \mathrm{U}(0, 1)$, a lattice observation $Y = \min\{\mathrm{round}(Y_0 + \varepsilon), 125\}$ with $\varepsilon \sim N(0, 9^2)$, and a deliberately biased predictor $\hat{y} = 1.06\, Y_0 + \eta + 4$ with $\eta \sim N(0, 5.5^2)$, left unclipped so that over-prediction near the cap produces the same one-sided residuals as C-MAPSS. Each configuration is repeated 200 times with a fresh test set of 4 000 engines.

V. Results

A. Width

Table I orders the constructions by mean width. Three regimes are visible.

The frozen 66-parameter layer is the sharpest object, at 30.9 cycles, and the shortest-interval rule applied to its own predictive distribution takes it to 27.7 — 10.4 % narrower at *identical* randomised coverage, because both intervals carry 0.90 of the same pmf. This is the first pure convention cost: the central interval is the customary summary of a predictive distribution, and on a censored RUL pmf it is not the cheapest summary of the same information.

Among conformal constructions, the matched-object family is 31.4–32.4 cycles and the CV+ family 35.3–36.3, an 11 % difference for the same guarantee. The gap is the fold spread: CV+ calibrates the model trained on $4/5$ of the data and deploys the average of five, so it must cover the difference. Split conformal with a single fold model, which is neither object, pays a further penalty (36.7–43.1) because its centre is a weaker predictor.

Between equal length and free tails there is no consistent winner on this benchmark (31.4 versus 32.2–32.4, i.e. within 3 %), for the reason given in Section III: the C-MAPSS residual law is sharply peaked, and the 90th percentile of $|R|$ sits below the 95th percentile of R. The sign reverses on data where the residuals are diffuse; Section V-E reports 19 % in favour of free tails there, and the simulation 15–19 %.

B. Coverage

The layer's coverage ranges 0.842–0.932 across the four fleets, a spread of 0.090. Two of the four fleets are below nominal by 0.03–0.06, and with $n = 228$ on FD004 that is 2.9 binomial standard errors: the layer is not merely noisy, it is wrong on two fleets and conservative on the other two. The conformal constructions compress the range to 0.026–0.058. EAC gives 0.897–0.925 with $|\Delta| = 0.014$; the equal-length split construction gives $|\Delta| = 0.009$ but pays 4.3 cycles of width for it.

No construction is uniformly best, and the table should be read as a frontier rather than a ranking: sharpness (layer-shortest, 27.7), calibration stability (split equal-length, 0.009), and the compromise between them (EAC, 32.4 / 0.014). The Winkler score prefers the layer (41.0), because its heteroscedastic shape concentrates width where the residuals are large, which is the one property the conformal constructions cannot borrow without a variance model.

The paired version of these comparisons, promised in Section II, changes what they say. The matched construction with a symmetric band and the same construction with a free tail carry the same 0.90 mass, so any gap between them is pure convention. On FD001 the two disagree on 11 of 279 engine-observations, every one of them in favour of the free tail (exact McNemar $p = 0.001$); on FD004 they disagree on 43 of 684, 30 to 13 the other way ($p = 0.014$). The same two rules are thus significantly better and significantly worse on different fleets at the same nominal level. FD002 (24 discordant, $p = 0.31$) and FD003 (4, $p = 0.13$) do not reject. A marginal table gives one number per fleet and hides the fact that the two rules err on different engines, which is the reason the positive increments in Table IV coexist with the negative ones.

C. Standard baselines

Table II places the constructions that are actually deployed next to the proposed one. Three are worth separating.

A fifteen-member deep ensemble scored as $\mu \pm z_{0.95}\sigma$ covers 0.485–0.553. At a nominal 90 % level it misses half the fleet, and the failure is structural rather than a tuning error: the members share their training data, so their spread measures seed variance and not predictive error, and averaging fifteen correlated models makes the mean much more accurate than the spread of its members. (Using five members instead of fifteen lowers coverage further, to 0.416–0.504, so the reported baseline is the stronger of the two.) Reporting that band as a 90 % interval understates the error by a factor of two, and since the same recipe is common in the RUL literature it is worth stating as a measurement result: the width of an uncalibrated ensemble band is not an uncertainty.

Quantile regression on the predicted RUL, fitted by linear programming on the same out-of-fold engines, reaches 0.739–0.809 on its own. The conformal offset is what buys the guarantee: CQR [15] reaches 0.883–0.919 at 32.8–41.6 cycles, i.e. the offset repairs 10–14 coverage points and costs 1.4–4.7 cycles of width relative to the uncalibrated quantile model. Both are fitted as a hinge basis in the predicted RUL with

TABLE I

INTERVAL CONSTRUCTIONS ON THE FOUR C-MAPSS FLEETS AT NOMINAL 0.90. COVERAGE IS THE EMPIRICAL FREQUENCY ON THE 653 TEST ENGINES, MEAN OVER THREE SEEDS; WIDTH IS IN CYCLES; $|\Delta|$ IS THE MEAN ABSOLUTE DEVIATION FROM 0.90 ACROSS FLEETS AND "SPREAD" IS THE DIFFERENCE BETWEEN THE BEST AND WORST FLEET. WINKLER IS THE INTERVAL SCORE AT $\alpha = 0.10$ (LOWER IS BETTER). THE FROZEN LAYER IS SCORED WITH RANDOMISED PIT, WHICH IS THE UNBIASED CONVENTION; ITS INTEGER-ENDPOINT SCORE IS 0.886–0.948.

Construction	FD001	FD002	FD003	FD004	$\lvert\Delta\rvert$	spread	width / Winkler
Frozen layer, central	0.925	0.869	0.932	0.842	0.037	0.090	30.9 / 41.0
Frozen layer, shortest	0.925	0.869	0.932	0.842	0.037	0.090	**27.7** / 43.7
Split, equal length	0.892	0.918	0.899	0.908	0.009	0.026	36.7 / 52.7
Split, free tails	0.900	0.938	0.932	0.927	0.024	0.039	43.1 / 55.1
CV+, equal length	0.914	0.943	0.893	0.943	0.027	0.050	35.3 / 48.9
CV+, free tails	0.903	0.928	0.893	0.934	0.018	0.041	36.3 / 49.7
EAC, equal length	0.885	0.922	0.883	0.942	0.024	0.058	31.4 / 45.8
EAC, equal-tailed	0.882	0.909	0.880	0.925	0.018	0.046	32.2 / 47.3
EAC (free tails)	0.925	0.913	0.897	0.917	**0.014**	**0.028**	32.4 / **45.2**

TABLE II

BASELINES A PRACTITIONER WOULD DEPLOY, AT NOMINAL 0.90, ON THE SAME 15 MODELS, THE SAME 100 CALIBRATION ENGINES AND THE SAME TEST ENGINES AS TABLE I. COVERAGE IS THE EMPIRICAL FREQUENCY, MEAN WIDTH IN CYCLES, $|\Delta|$ THE MEAN ABSOLUTE DEVIATION FROM 0.90 ACROSS FLEETS; ALL THREE SHARE THE ENSEMBLE CENTRE $\bar{\mu}$ AS THE POINT PREDICTION AND DIFFER ONLY IN THE INTERVAL RULE.

Method	FD001	FD002	FD003	FD004	$\lvert\Delta\rvert$	width	Winkler
Deep ensemble $\pm z_{0.95}\sigma$, no offset	0.548	0.528	0.485	0.553	0.372	12.6	74.7
Quantile regression, no offset	0.808	0.809	0.739	0.787	0.114	34.1	50.0
CQR [15]	0.909	0.919	0.883	0.890	0.014	37.2	48.1
EAC (free tails)	0.925	0.913	0.897	0.917	0.015	32.4	45.2

two knots and cross-fitted over two halves of the calibration engines, so that no engine is used both to fit the quantile function and to set the offset; a quadratic basis was tried first and rejected because the resulting collinearity made the linear programme return a degenerate vertex (intervals 30 % too narrow at nominal 0.90).

Against that baseline the matched construction is 14–22 % narrower at equal or better coverage on three of four fleets, and level on FD004 (41.9 against 41.6, where the fleet's six operating conditions put most of the width in the same place). Fleet-averaged stability is a tie ($|\Delta| = 0.015$ against 0.014): CQR is a strong baseline on calibration and the difference between the two is sharpness, not validity. At nominal 0.50 CQR returns 0.393–0.556 against 0.392–0.512 for EAC: both are far from 0.50, for the reason discussed in Section VI, and neither should be read as a validated 50 % statement at this sample size.

D. Level consistency

A convention that pays at one nominal level only is not a convention. Table III repeats the two width comparisons at four levels.

The convention cost of the tail split is monotone in the level: 32.9 % at 0.50, 15.1 at 0.80, 10.3 at 0.90 and 7.9 at 0.95, with the fleet spread shrinking from 6.2 to 1.3 points over the same range. The mechanism is the cap. Once $1 - \alpha$ is wide enough that the central interval must reach the atom at 125 anyway, where the remaining mass is placed matters less, and the quadruple $(\gamma, \alpha - \gamma)$ buys correspondingly less. Quoting the cost at 0.90 alone therefore hides a factor of three against the 0.50 interval.

The object-matching gain has no such trend: 20.3 % at 0.50, 14.1 at 0.80, 7.9 at 0.90 and 8.1 at 0.95, positive on all four fleets at the two tighter levels and negative on one fleet at each of the two looser ones. The numerator of that ratio is the fold spread, which is set in cycles by the training set, and it does not grow with the residual quantile the way the denominator does.

At 0.95 the ordering against the parametric layer also changes, and this is the one place where the table should not be read as a win. The layer is narrower on three of four fleets (35.9–36.1 cycles against 40.5–48.2) while covering 0.901–0.973 against 0.932–0.983; on FD004 it is 12.1 cycles narrower and 4.9 coverage points short of the level. The narrowest intervals in the table at a nominal 0.95 belong to the construction that misses that level by the widest margin, which is the paper's argument in a single comparison.

At nominal 0.50 the ranking changes again. The layer returns 0.496 on average (0.517/0.523/0.524/0.420); CV+ returns 0.560 (up to 0.638 on FD002) because the fold-spread penalty is fixed in cycles and therefore relatively larger on a 12-cycle interval; EAC returns 0.464, with FD003 at 0.392 as its worst case. On this benchmark no conformal construction is reliable at 50 %, and we report that as a limit rather than tuning it away: with 93–235 engines the q_{25} and q_{75} of the residual law carry sampling error of ± 0.03–0.05 in coverage, and the two populations differ by more than that (Section VI). That gap is a covariate shift; weighted conformal [16] and the non-exchangeable bound of [14] say what a guarantee becomes when it is present. We use the bound as the explanation of the fleet-to-fleet variation rather than as a correction, because the weights cannot be estimated reliably at 100 engines.

TABLE III

RELATIVE WIDTH CHANGE AT FOUR NOMINAL LEVELS, IN PERCENT; POSITIVE MEANS THE SECOND CONSTRUCTION IS NARROWER. LEFT: THE PARAMETRIC LAYER'S CENTRAL INTERVAL AGAINST ITS SHORTEST INTERVAL OF THE SAME MASS — THE PURE CONVENTION COST, AT IDENTICAL RANDOMISED COVERAGE. RIGHT: THE MATCHED CONSTRUCTION AGAINST CV+ AT THE SAME LEVEL — THE OBJECT-MATCHING GAIN. ENTRIES ARE PER-FLEET RATIOS; THE FLEET-MEAN RATIO GIVES 10.4 % AT 0.90, AGAINST 10.3 FOR THE MEAN OF THE FOUR RATIOS BELOW.

	central → shortest, layer				matched against CV+			
Level	FD001	FD002	FD003	FD004	FD001	FD002	FD003	FD004
0.50	29.6	30.3	35.9	35.8	15.1	19.1	25.1	21.8
0.80	13.8	14.2	15.8	16.7	21.9	18.4	7.9	8.1
0.90	9.4	10.1	10.3	11.5	17.3	21.3	0.8	−7.9
0.95	7.4	7.9	7.5	8.7	18.9	18.3	−5.3	0.3

E. N-CMAPSS

On the independent benchmark, with leave-one-fleet-out calibration and a bootstrap over the 39 test units, the matched construction returns coverage 0.896 (95 % interval $[0.840, 0.942]$) at nominal 0.90 and 0.493 ($[0.429, 0.557]$) at nominal 0.50, with widths 39.0 and 13.2. The equal-length construction returns 0.945 and 0.750 at widths 48.4 and 25.8: it over-covers by 0.05 and 0.25 while being 19 % and 49 % wider. This is the clearest evidence in the paper for the free-tail construction, and it comes from data where the RUL cap is inactive, so it is not an artefact of the C-MAPSS censoring convention.

F. Simulation

With known truth (Fig. 2), both families are valid: the equal-length band returns 0.899–0.902 at nominal 0.90 and 0.508–0.510 at nominal 0.50, and the free-tail band 0.899–0.906 and 0.505–0.509. The difference is width: 31.3 versus 37.4 cycles at 0.90 (16 % narrower) and 13.1 versus 16.6 at 0.50 (21 % narrower). The tail allocation γ chosen on the same sample returns 0.898 and 0.501, i.e. the selection costs 0.002–0.005 of coverage, and saves a further 0.1–0.2 cycles.

The simulation also fixes the interpretation of the real-data result. Because the generator's residuals are diffuse, free tails win by 16 %; because C-MAPSS residuals are peaked, they do not. Neither outcome is a property of conformal prediction. Both are properties of the score law, and the honest procedure is to look at it.

G. Decision consequences

At a planning horizon of 30 cycles, an engine is flagged when its upper bound falls inside the window. On FD003 the late-alarm rate (true RUL < 30 and not flagged) is 0.103 for EAC and 0.096 for the equal-length band; on FD004, 0.187 versus 0.228 at the same horizon. The differences track the upper endpoint, not the coverage: a construction that is wider on the upper side flags earlier and is late less often, at the cost of flagging engines that did not need it. We report the pair rather than a single cost because the trade is a policy choice.

H. Ablations

Table IV removes one ingredient at a time from EAC and leaves the other three in place, recomputing the free tail inside each variant because the allocation is part of that variant's definition. The metric is the Winkler interval score at nominal 0.90, which is the joint criterion: width alone would rank the symmetric band first (31.3 cycles mean against 32.4) even though it scores worse on the interval score, because it puts its mass on the wrong tail. Each entry is the change relative to EAC, positive meaning the removal hurts, with an engine-level block bootstrap 95 % CI.

Variance normalisation is the ingredient that pays: removing it costs $+5.0$ Winkler points on average, $+10.4$ on FD001, and the CI excludes zero on three of four fleets. Object matching is second ($+0.9$ on average) and is the one that matters on the two fleets where the fold spread is small relative to the ensemble residual (FD001 $+4.3$, FD003 $+4.2$, both CIs excluding zero). The sign and the tail allocation matter only on FD001. No removal is significantly better than EAC on any fleet: all seven negative increments have CIs that include zero, and the largest of them, -3.1 on FD004 for object matching, means that on a fleet with six operating conditions the fold-matched construction is as good, not better.

The construction costs 0.02 ms per query on 100 calibration scores and 228 test engines, against seconds for the backbone forward pass, so the interval layer is free in deployment. Two variants were tried and are not used: replacing the two-parameter variance function with a binned median-absolute-deviation scale is worse on three of four fleets, and replacing the paired split set with the union over all fold–residual pairs changes the width by -4 to $+6$ % with no coverage gain.

The free tail deserves a separate note because it is the only ingredient whose value is chosen from data rather than fixed by construction. In the simulation, where calibration and deployment are exchangeable, selecting γ on the calibration sample costs 0.002 of coverage at $m = 2000$ (0.8984 against 0.9006 for equal tails), and a five-fold choice inside the calibration set, scored by the out-of-fold interval score, returns 0.9005. On the benchmark the same rule is 5.0 Winkler points worse on average than the in-sample choice, and it disagrees with it by up to 0.078 of the admissible range of γ (0.012 against 0.090 on FD001): calibration and deployment are not exchangeable there, and the allocation is precisely the degree of freedom that reads the difference. The free tail is therefore reported as an upper bound, and the ablation above, which never sees the test set, as the lower one.

TABLE IV
LEAVE-ONE-OUT ABLATION OF EAC AT NOMINAL 0.90. EACH COLUMN REMOVES ONE INGREDIENT AND RECOMPUTES THE REST, INCLUDING THE TAIL ALLOCATION. ENTRIES ARE THE CHANGE IN WINKLER INTERVAL SCORE RELATIVE TO THE FULL CONSTRUCTION, POSITIVE MEANING THE REMOVAL HURTS, WITH A 95 % ENGINE-LEVEL BLOCK BOOTSTRAP CI IN BRACKETS. REFERENCE ROW: EAC ABSOLUTES. THE SYMMETRIC BAND IS 3 % NARROWER ON AVERAGE YET SCORES WORSE, BECAUSE WIDTH IS NOT THE OBJECTIVE IT IS OFTEN TAKEN TO BE.

Removed	FD001	FD002	FD003	FD004	mean
EAC absolute: coverage / width / Winkler at nominal 0.90					
	0.925/28.0/36.4	0.913/31.9/44.2	0.897/27.9/43.0	0.917/41.9/57.3	
object matching	$+4.3\,[2.4,\ 6.3]$	$-1.8\,[-4.2,\ 0.2]$	$+4.2\,[0.2,\ 8.1]$	$-3.1\,[-7.7,\ 1.4]$	$+0.9$
variance normalisation	$+10.4\,[4.6,\ 17.4]$	$+7.4\,[5.1,\ 9.5]$	$+2.8\,[1.7,\ 4.0]$	$-0.6\,[-2.2,\ 1.2]$	$+5.0$
the sign	$+5.1\,[0.4,\ 10.6]$	$-1.4\,[-3.8,\ 0.7]$	$+1.3\,[-0.6,\ 3.3]$	$-2.5\,[-7.5,\ 2.3]$	$+0.6$
the free tail	$+9.5\,[2.1,\ 17.2]$	$-1.2\,[-4.2,\ 1.5]$	$+2.6\,[-0.2,\ 5.5]$	$-2.6\,[-5.3,\ 0.3]$	$+2.1$

I. A second backbone

Every number so far comes from one network. To separate the conventions from the model, the whole construction set was recomputed on a second backbone: a two-layer bidirectional LSTM with additive attention pooling, 119 490 parameters against 41 858, trained on the same engine folds with the same fold-wise scalers, the same deployment cut-offs and the same three seeds. It is a weaker predictor — dev RMSE 13.1/15.3/12.7/15.4 against 11.1/13.9/10.5/13.9 — which makes it a harder test, not an easier one.

Two things replicate and one does not. The width ordering replicates on all four fleets: EAC is 9.1, 17.9 and 10.0 % narrower than CV+ on FD001–FD003 and 5.4 % wider on FD004, the same sign pattern as the primary backbone (17.2, 21.4, 0.7 and -8.0 %). The gap to the parametric layer replicates too, and widens: on the second backbone the layer covers 0.803–0.905 at widths of 38.4–62.8, and EAC is 4–37 % narrower than it.

What does not replicate is the coverage level. On the same engines, the same rules and the same nominal level, coverage falls to 0.766–0.925 for EAC and 0.814–0.940 for CV+, with the deficit concentrated on FD003, where the second backbone's own parametric layer covers 0.803. The deficit is a property of the predictor–population pair, not of the interval rule: a weaker network has larger residuals on deployment inputs than on the held-out training engines, and no marginal construction can see that shift. This is the paper's thesis in controlled form — two predictors, one set of rules, a 0.13 swing in coverage and a stable width ordering — and it is why Table I reports width next to coverage rather than instead of it.

VI. DISCUSSION

A. Why coverage tables cannot rank methods at this scale

Fig. 1 makes the argument visually. At $n = 93$ the 95 % band around a true 0.90 is ± 0.06; at $n = 235$ it is ± 0.04. Every construction in Table I except the layer lies inside the band on at least three of four fleets, and two constructions that differ by 30 % in width are indistinguishable on coverage. A table that reports 'PICP 0.912 versus 0.907' is therefore reporting sampling noise, not evidence, unless the comparison is paired and the discordance is reported.

The detectability calculation in Table VII makes the same point quantitatively and in advance of any experiment: with 11–82 engines per bin, the smallest detectable deviation is 0.14–0.44. The five-bin comparison that a conditional-coverage table implies requires 149 engines per bin for a 10-point effect, and 516 for a 5-point effect. Pooling all four fleets gives 84–212 per bin, which is enough for a 0.08–0.14 effect on the most populated bin and nothing on the least.

B. What to report

We suggest four rules, each of which is a measurement fix rather than a modelling choice. (i) Report coverage with its interval, and report the number of engines, not only the rate. (ii) Use randomised PIT for distributions; the integer rule is biased upward by 0.01–0.04 at 0.90 and 0.07–0.12 at 0.50. (iii) Report width, which is measurable to ± 1 cycle and separates methods that coverage cannot, and report it per fleet, because the fleet is a larger source of variation than the method ($|\Delta|$ 0.09 across fleets for the layer versus 0.02 between constructions). (iv) State which object was calibrated. If the deployed predictor is an ensemble, a single-model conformal score is not the same object, and the difference is 11 % of the width.

C. Limitations

The study covers two simulated turbofan benchmarks and two backbones. Both datasets are run-to-failure simulations, so the residual laws may be gentler than those of instrumented machinery; the N-CMAPSS contrast suggests the direction of the difference (diffuse residuals, where free tails win by 19 %) but does not establish it for real sensor data. The 50 % level is not resolved on C-MAPSS by any construction, and we have not found a construction that resolves it; we report the failure rather than selecting the seed that hides it. The free-tail allocation is the least reliably estimated of the four ingredients: a five-fold choice inside the calibration set disagrees with the in-sample choice by up to 0.078 of the admissible range of γ and is 5.0 Winkler points worse on average, because the calibration and deployment RUL mixtures differ. We therefore report the in-sample allocation as an upper bound on what the free tail can buy, and the ablation of Table IV — which is independent of the test set — as the evidence that each ingredient is non-negative. Finally, the guarantees are marginal, and the paper's own detectability analysis is the reason we do not claim conditional ones.

TABLE V
SAME CONSTRUCTIONS, SECOND BACKBONE, NOMINAL 0.90. COVERAGE / MEAN WIDTH IN CYCLES. THE SECOND BACKBONE IS 2.9 TIMES LARGER AND 10–21 % WORSE IN DEV RMSE; THE WIDTH ORDERING OF THE CONSTRUCTIONS IS UNCHANGED, WHILE THE COVERAGE LEVEL MOVES BY UP TO 0.13. THE PARAMETRIC LAYER IS REFITTED ON EACH BACKBONE'S OWN RESIDUALS.

Backbone		FD001	FD002	FD003	FD004
primary (41 858)	CV+	0.914 / 33.8	0.943 / 40.6	0.893 / 28.1	0.943 / 38.8
	EAC	0.925 / 28.0	0.913 / 31.9	0.897 / 27.9	0.917 / 41.9
second (119 490)	layer	0.866 / 38.4	0.905 / 56.0	0.803 / 47.4	0.904 / 62.8
	CV+	0.907 / 40.5	0.935 / 44.2	0.814 / 33.1	0.940 / 44.3
	EAC	0.875 / 36.8	0.901 / 36.3	0.766 / 29.8	0.925 / 46.7
EAC vs CV+, width		−17.2 / −9.1	−21.4 / −17.9	−0.7 / −10.0	+8.0 / +5.4

top row of each pair: primary; second row: second backbone. The last line gives (primary / second) relative width change.

TABLE VI
COVERAGE BIAS OF THE INTEGER-ENDPOINT RULE ON THE FROZEN LAYER, AND THE SMALLEST COVERAGE DEVIATION A BIN CAN DETECT. LEFT: SAME PREDICTIVE DISTRIBUTION, TWO CONVENTIONS. RIGHT: $\delta^\star$ AT FAMILY-WISE $\alpha = 0.05$ OVER FIVE BINS AND 80 % POWER.

	bias at 0.90		bias at 0.50	
Fleet	PIT	integer	PIT	integer
FD001	0.924	0.935	0.517	0.591
FD002	0.869	0.902	0.523	0.643
FD003	0.931	0.948	0.524	0.608
FD004	0.842	0.886	0.420	0.535
bias	+0.011 to +0.044		+0.074 to +0.120	

TABLE VII
WHAT A BIN CAN PROVE. BIN SIZES ARE THE FIVE RUL STRATA OF THE TEST SETS; $\delta^\star$ IS COMPUTED AT FAMILY-WISE $\alpha = 0.05$ OVER FIVE BINS WITH 80 % POWER.

	engines per bin				
Fleet	$[0, 25)$	$[25, 50)$	$[50, 75)$	$[75, 100)$	$[100, 125]$
FD001	19	11	13	22	28
FD002	58	29	31	45	72
FD003	15	14	20	18	30
FD004	49	30	26	41	82
pooled	141	84	90	126	212

$\delta^\star$ at $n = 11$: 0.44 $n = 82$: 0.14 $n = 141$: 0.10 $n = 212$: 0.08
engines needed: 149 for $\delta^\star = 0.10$, 243 for 0.075, 516 for 0.05

VII. CONCLUSION

RUL intervals are reported as a coverage and a width, and this paper has separated the part of each that is information from the part that is convention. On 653 C-MAPSS engines and 2 587 N-CMAPSS queries: the integer-endpoint convention inflates coverage by 0.011–0.044 at 0.90; the central-interval convention costs 10.4 % of width, and three times that at the 50 % level; calibrating a fold model instead of the deployed ensemble costs 11 %; and the benchmarks can detect only 0.14–0.44 deviations per bin, which is larger than the differences that published tables rank. A matched-object construction recovers the layer's sharpness (32.4 versus 30.9 cycles) with three times its fleet stability ($|\Delta|$ 0.014 versus 0.037) and without its distributional assumptions, and reaches nominal coverage on the independent benchmark where the parametric layer does not. Its four ingredients are individually non-negative under leave-one-out ablation, and its width ordering survives transplantation onto a second backbone, where coverage does not: on the same engines and the same rules, a weaker predictor moves coverage by up to 0.13 while the ordering of widths is preserved. The broader conclusion is that the field's uncertainty numbers are now limited by measurement practice more than by modelling: the same 653 engines that cannot resolve a 3-point coverage difference can resolve a 10 % width difference, and can resolve it even when the predictor changes underneath, so width and its conventions are where the next improvements will be visible.

APPENDIX A
PROOF OF PROPOSITION 1

Assume continuous scores, so that ties have probability zero; with ties, randomise the rank. Let $R = \#\{i \le m : S_i \le S\}$. Under exchangeability of $(S_1, \dots, S_m, S)$ the rank of S among the $m+1$ values is uniform, so $\Pr[R = r] = 1/(m+1)$ for $r = 0, \dots, m$.

(i) $S < L_\gamma$ holds iff fewer than $\lfloor (m+1)\gamma \rfloor$ calibration scores lie below S, i.e. iff $R \le \lfloor (m+1)\gamma \rfloor - 1$, hence $\Pr[S < L_\gamma] \le \gamma$. Likewise $S > U_\gamma$ iff $R \ge \lceil (m+1)(1-\alpha+\gamma) \rceil$, hence $\Pr[S > U_\gamma] \le \big(m+1-\lceil (m+1)(1-\alpha+\gamma) \rceil\big)/(m+1) \le \alpha - \gamma$. The two events are disjoint, so the miss probability is at most α for every γ.

(ii) With $k = \lceil (m+1)(1-\alpha) \rceil$ and $\gamma = j/(m+1)$, (3) reads $[s_{(j)}, s_{(j+k)}]$: as j ranges over the admissible grid the family is the set of windows of $k+1$ consecutive order statistics, and the minimiser of $s_{(j+k)} - s_{(j)}$ is found by scanning j after one sort, which is $O(m \log m)$ and is what the implementation does. In the population limit $U - L = q_{1-\alpha+\gamma} - q_\gamma$, whose stationarity condition is $f(q_{1-\alpha+\gamma}) = f(q_\gamma)$.

(iii) Substituting $\gamma = \alpha/2$ gives $f(q_{\alpha/2}) = f(q_{1-\alpha/2})$, which holds for a symmetric unimodal law and fails otherwise. For the capped RUL law the upper quantile is the atom at the cap, where no density exists, so the condition fails by construction. The attainable gain is the width of the central interval minus the width of the shortest window, which Table I measures at 3.2 cycles (30.9 against 27.7) for the parametric layer and Table IV at 2.1 Winkler points for the conformal construction.

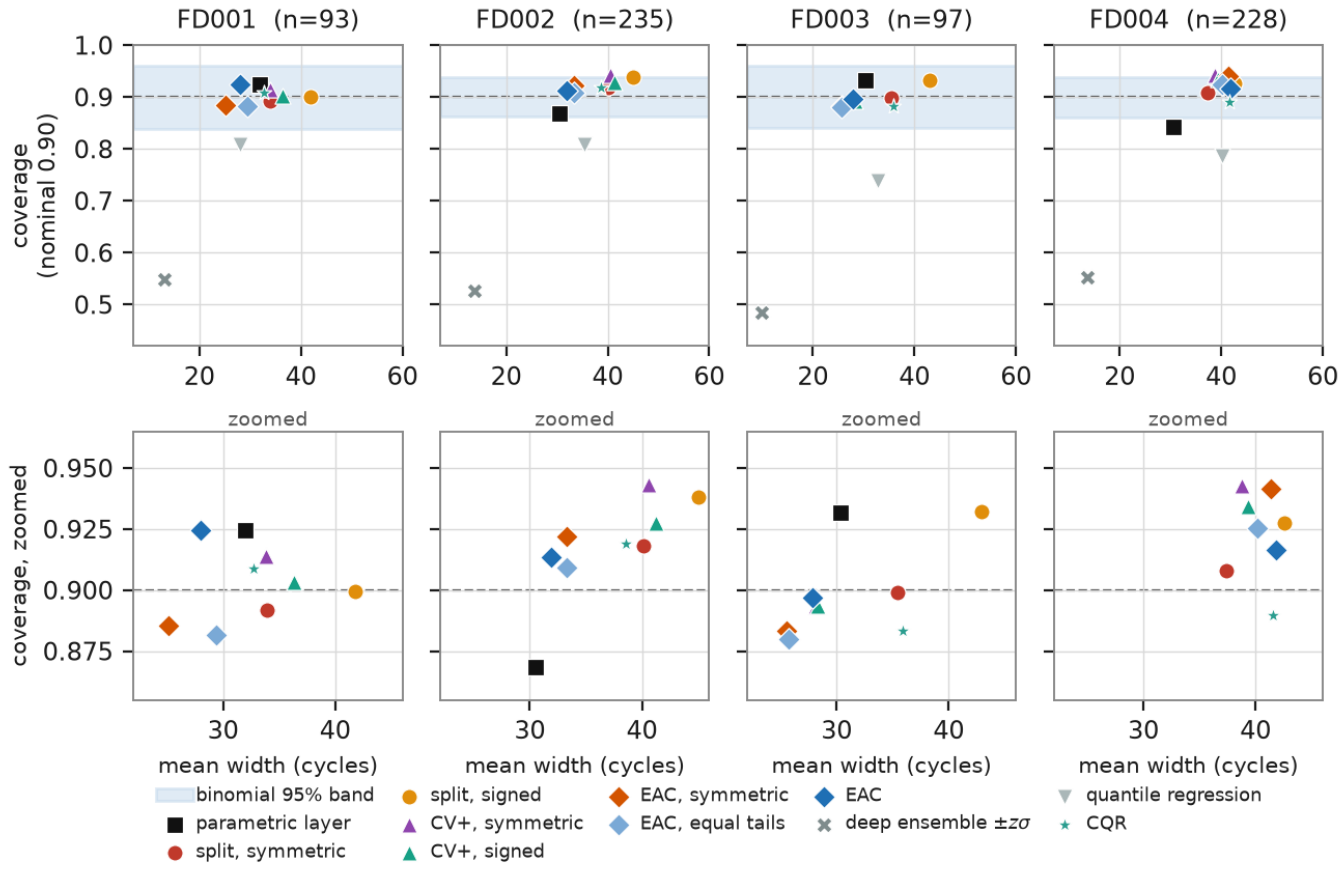


Fig. 1. Coverage against width at nominal 0.90, one panel per fleet. Top row: the full range, including the two constructions that need no calibration (grey); the shaded band is the binomial 95 % interval for a true coverage of 0.90 at that n. Bottom row: the same points with the axes narrowed to the conformal cluster. Uncalibrated ensemble spread sits half a fleet below nominal at a third of the width, and the calibrated constructions sit inside the band: coverage cannot separate them at this scale, while width separates them by 30 %.

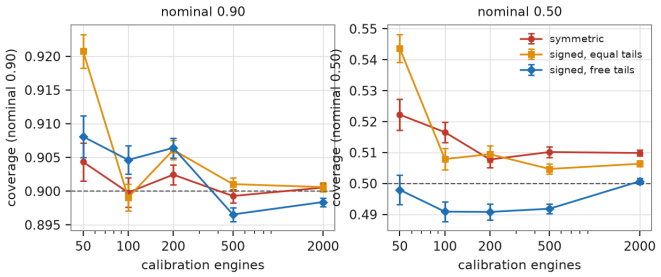


Fig. 2. Simulation with known truth: coverage against the number of calibration engines, at nominal 0.90 (left) and 0.50 (right). All three constructions are valid; they differ in width by 16–21 % (not shown).

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